Ammar Hussain

PhD Candidate | University of Houston | 3507 Cullen Blvd | TX 77204

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PROFESSIONAL SUMMARY

Spectral Geologist & GIS Specialist with 4 years of experience in Hyperspectral and Multispectral analysis, geospatial analysis, UAV mapping, satellite and insitu spectral mapping, and spatial modeling for mining, infrastructure, and environmental projects. Skilled in ENVI 6.2, Hyperspectral Image Processing and Interpretation, ArcGIS Pro, and Python, with strong experience translating geospatial remote sensing data into actionable insights

Education

Ph.D. | Geology **|** Department of Earth and Atmospheric Sciences, University of Houston **|** cgpa (3.857/4) Graduation: May 2027

Thesis: Multi-Scale Detection and Exploration of Critical Minerals Based on Remote Sensing Techniques

M.Phil. | Geology **|** Institute of Geology, University of Punjab

Thesis: Rock Mass classification of the headrace tunnel of a small hydropower project, Kandian Valley, Pakistan

Publication

- **Hussain**, Ammar and Shuhab Khan. "Multiscale Hypersepctral Remote Sensing and Integrated Geochemical Analysis for Pegmatites Hosted Critical Minerals" (Under Preparation)
- **Hussain**, Ammar, Shuhab Khan, and Jagos R Radovic. "Hyperspectral Imaging of Coal Ash for Rare Earth Elements" Fuel, Volume 409, 2026, 137676, ISSN 0016-2361, https://doi.org/10.1016/j.fuel.2025.137676.
- **Hussain**, Ammar, Amna Afzal, and Shuhab D. Khan. "Tectonic analysis of the Mula River Basin, Kirthar fold belt, Pakistan, using hypsometric index." *Interpretation* 12, no. 2 (2024): SC9-SC16.

(Complete publication list can be provided on request)

Experience

Research Experience – Earth and Atmospheric Sciences Department, University of Houston, August 2022 – Present

Project # 1: Critical Mineral exploration of Shigar Valley based on field data and hyperspectral Imaging.

- Satellite-based Spectral Analysis of the region to detect mineral occurances.
- Field work and sampling of pegmatite samples in the Karakoram Ranges.
- Samples were analyzed using a spectroradiometer (ASD Fieldspec Pro) and hyperspectral cameras (VNIR, SWIR and LWIR) to map minerals based on Spectra.
- LIBS, Elemental Analyzer, and pXRF testing of the samples have been completed to validate our studies.

Project # 2: Detection of Rare Earth Elements in Coal and Fly Ash Samples through Hyperspectral Imaging Techniques

- Coal and fly ash samples preparation and hyperspectral imaging (VNIR-SWIR) to develop a remote sensing methodology in detecting critical minerals.
- XRD studies of coal samples were conducted to identify mineral composition.
- Bulk rock geochemistry using pXRF and hyperspectral studies completed.

Professional Experience – Geologist

I have worked as a Geologist on multiple multidisciplinary projects involving water resource management, hydropower development, and mineral exploration. Contributed to copper exploration campaigns through UAV and Satellite-based orthoimagery, geological mapping, sampling, and data interpretation. Additionally, participated in geohazard assessments, integrating geological and geotechnical investigations to support safe and sustainable infrastructure development.

Relevant Ph.D. Coursework

(1) Remote Sensing (2) Geospatial Analysis (3) Deep Learning for Big Data Analytics (4) Satellite Positioning and Geodesy (5) Paleoclimate (6) Planetary Materials (7) Seafloor Mapping

Skills

Geological Remote sensing, Hyperspectral Imaging, Geospatial Analysis, Hyperspectral & Multispectral data processing, Mineral Exploration, pXRF analysis, XRD analysis, and Slope stability studies.

Software and Programming

ArcGIS Pro 3.2, Envi 6.0 Software, RocScience Slide2, Python language, Deep learning & Machine learning.

Conference Presentations

- **Hussain**, Ammar, and Shuhab Khan. "Hyperspectral Imaging for Rare Earth Elements of Coals from Navajo Mine, New Mexico." In Geological Society of America Abstracts, vol. 56, p. 404793. 2024.
- **Hussain,** Ammar, Shuhab Khan, Jagos Radovic. "Hyperspectral Imaging and Analysis of Coal Ash for Rare Earth Elements." Abstract no. V23B-0097 presented at AGU25, New Orleans, LA, 15-19 Dec.
- Radovic, Jagos, Thomas Malloy, Tao Sun, Shuhab Khan, Yongjun Gao, John Casey, and **Ammar Hussain**. "Exploring Critical Elements in Unconventional Sources Using Hyperspectral and Geochemical Methods." *Authorea Preprints* (2025).

Workshops

- North American Workshop on Critical Mineral Research, Development and Education, Austin, Texas, USA
- EarthCube Advancing the Analysis of High Resolution Topography (A2HRT) workshop, Tempe, Arizona, USA

Achievements

- Student Research Grant – Geological Society of America 2025
- Travel Grant for North American Workshop on Critical Mineral Research, Development, and Education, Austin, Texas, USA
- Earth and Atmospheric Science Scholarship for Outstanding Graduate Work in Geology Award - University of Houston 2025
- Earth and Atmospheric Science Outstanding Graduate Research Award - University of Houston 2024
- Travel grant from GSA for presentation at Connect 2024 in Anaheim, CA, USA
- Earth and Atmospheric Science Outstanding Graduate Research Award - University of Houston, Fall 2023
- Travel Scholarship for participation in the EarthCube Advancing the Analysis of High Resolution Topography (A2HRT) workshop, Tempe, Arizona, USA